Abishek Soti
linkedin.com/in/abisheksoti |Google Scholar | github.com/AbishekSoti

Education

Master of Philosophy Western Sydney university| August 2025 - Current

- **Fully funded competitive research scholarship (tuition + stipend)**
- Research focus: neuromorphic sensing, event-based vision, underwater acoustics, ML algorithms.
- Published a paper on Dronebased sound source localisation as a co-author(2025).

Bachelor of Engineering (Honours) in Electrical Engineering Western Sydney University | July 2025

- Relevant coursework: signal processing, biomedical signal analysis, digital signal processing, embedded systems.

Experience

M.Phil. Researcher – Underwater Acoustic Anomaly Detection
International Centre for Neuromorphic Systems, MARCS August 2025- Present (Full time)

- Designed Python-based ML pipelines for audio classification, supporting dataset processing, feature extraction, model training, and evaluation across CNNs, Transformers, and classical ML models.
- Increased classification accuracy from approximately **75% baseline to 98%** by redesigning audio feature representations using spectral transformations and statistical pooling.
- Improved class separability across ShipsEar and DeepShip using CQT, MFCC, Mel features, and statistical feature engineering.
- Implemented and evaluated CARFAC auditory front-end processing for biologically inspired signal representation in neuromorphic ML systems.

Research Assistant |International Centre for Neuromorphic Systems, MARCS September 2024- Present

- Designed and executed event camera characterisation experiments to detect sensor anomalies (e.g., hot/cold pixels), improving reliability of downstream data interpretation
- Developed Arduino-based control algorithms to generate logarithmic light stimuli for sensor calibration experiments
- Built Python pipelines to process asynchronous event streams and extract temporal features from neuromorphic data
- Integrated offline (Faery) and live sensor pipelines for event-driven data acquisition and analysis

Technical Skills

Languages: Python, SQL, MATLAB, JavaScript, HTML/CSS, VHDL/Verilog
ML/Data: PyTorch, scikit-learn, CNNs, Transformers, LLM workflows, NumPy, pandas, Jupyter, model evaluation, hypothesis testing
Deployment/Tools: Git/GitHub, Hugging Face, OOP, technical documentation
Signal Processing: Audio classification, spectrograms, MFCC, CQT, Mel features, neuromorphic sensing

Projects

Audio Classification – UrbanSound8K, ShipsEar & DeepShip

- Built and deployed Python audio classification pipelines using CQT, MFCC, Mel/STFT features, statistical pooling, and supervised ML models; achieved **95% accuracy on UrbanSound8K with CQT + RBF-SVM** and improved DeepShip beyond baseline results with **92% RBF-SVM** and **93% scikit-learn MLP**.

FoodVision Image Classification – EfficientNet-B2

- Fine-tuned and deployed a pretrained EfficientNet-B2 model in PyTorch for FoodVision-style image classification, achieving approximately **95% accuracy** using transfer learning, DataLoaders, image transforms, and structured evaluation workflows.

NLP Job Description Classifier

- Built an NLP classification pipeline using TF-IDF, sentence-transformer embeddings, scikit-learn classifiers, and LLM-assisted zero/few-shot methods, comparing accuracy, macro-F1, confusion matrices, latency, and cost trade-offs.

Leadership & Activities

Western Sydney University | Student Ambassador, Master of Ceremonies, Treasurer | March 2023 – March 2025

- Supported student engagement events, orientation programs, and peer support initiatives across the Sydney City campus.